
C HANDOUT PROGRAMS

C-Language

- **SAMPLE PROGRAMS**

1. Write a program for addition, subtraction, multiplication, division. of two numbers.
2. Write a program to check whether a number is odd or even.
3. Write a program to find whether the number is perfect or not.

Write a program to swap to numbers without using third variable

- **Programs on conditional statements and loops**

1. Write a program to find a year is leap or not.
2. Write a program to check a char is vowel or not.
3. Write a program to reverse a number.
4. Write a program to check a number is palindrome or not.
5. Write a program to find factorial of a number.
6. Write a program to convert binary to decimal.
7. Write a program to convert decimal to binary.
8. Write a program to check a number is Armstrong or not.
9. Write a program to print the pattern.

1 2 3

1 2 3

1 2 3

When number=3



360digrii

10. Write a program to print the pattern

1 1 1

2 2 2

3 3 3

4 4 4

When number =4

11. Write a program to print the following pattern

*

* *

* * *

* * * *

When number =4

12. Write a program to print the following pattern

*

* * *

* * * *

* * * * *

13. Write a program to print this pattern

*

14. Write a program to print a number is prime or not.

15. Write a program to print 'n' prime numbers.

16. Write a program to print 'n' Fibonacci series.

ARRAYS

17. Write a program to find a length of an array.

18. Write a program to search the number in an array and return the index of that number.

19. Write a program to find maximum number in an array.

20. Write a program to find lowest element in an array.

21. Write a program to sort the array in an ascending order .

22. Write a program to sort the array in descending order

STRINGS

23. Write a program to read a string from key board.

24. Write a program to find reverse of a string.

25. Write a program to read a dynamic string from a key board and whether it is palindrome or not.

26. Write a Program to check a given character is present or not in a string.

27. Write a program to find the length of a string.



CORE JAVA PROGRAMS

Java programs

1. Write a program to check whether a number is odd or even.
2. Write a program to find whether the number is perfect or not.
3. Write a program to swap two numbers without using third variable

Programs on conditional statements and loops

1. Write a program to reverse a number.
2. Write a program to check a number is palindrome or not.
3. Write a program to find factorial of a number.
4. Write a program to convert binary to decimal.
5. Write a program to convert decimal to binary.
6. Write a program to check a number is Armstrong or not.
7. Write a program to print the following pattern

1

1 2

1 2 3

1 2 3 4

When number =4

8. Write a program to print the following pattern

```
  *
 * *
* * *
* * * *
```

9. Write a program to print this pattern

```
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

14. . Write a program to print 'n' prime numbers.

16. Write a program to print 'n' Fibonacci series.

CONSTRUCTORS, INSTANCE VARIABLES, STATIC VARIABLES

Write a program on method overloading and method overriding .

Write a sample code to show how to use constructors, and also how to initialize instance variables.

COLLECTION

1. Write a program to find the size of the linked list and return its size.
2. Write a program to add elements in an array list.
3. Write a program to add elements in a hash map.
4. Write a program to get the size of a linked hash map.
5. Write a program to get the element one by one in vector.
6. Write a program to get the elements one by one in a map.
7. Write a program to print the prime numbers and print it in reverse

order by using collections.

8. Write a program input=110010

MULTITHREADING

1. How to create a thread **in** java. Write a sample code.
2. Create 10 threads in java by a sample code.
3. Create a thread inside a thread if inner thread aborted the entire thread should stop executing.
4. Write a program which shows altering order of threads by changing thread priority.

EXCEPTION

1. How to create our own exception. Write a sample code.
2. How to handle unchecked exception with an example. Write a sample code

ADV JAVA PROGRAMS

JDBC

- 1) Write a JDBC Program to create a connection with database and display the connection object.
- 2) Write a JDBC Program to create a registration Table with following fields (Username, password, email, gender, date of birth, mobile number).
- 3) Write a JDBC Program to insert the values into the Registration table dynamically.
- 4) Write a dynamic JDBC Program to delete the record from registration table by using username.

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- 5) Write a JDBC Program to update a value in the registration table.
 - 6) Write a JDBC Program to get all the records from the registration table.
 - 7) Write a JDBC Program to find the number of records present in a table.
 - 8) Write a JDBC Program to get the records in sorted order.
 - 9) Write a JDBC Program to insert image into database.
 - 10) Write a JDBC Program to get the image from database.

SERVLETS

1. Write a Program to create a servlet using servlet interface and display the following output. Output: This is my first servlet using servlet interface.
2. Write a servlet program to create a connection to database.
3. Write an HttpServlet program to eliminate hard coding by using servletconfig or application object.
4. Write an HttpServlet program to get the values from html pages.
5. Write an HttpServlet program to get the values from the registration html page and insert into database.
6. Write an HttpServlet program to get the values from the database and display on the webpage.
7. Create an application of product using HttpServlets which has the following features
 - i) Html page to enter the product details (product id, product name, product price).
 - ii) Servlet to get the product details from the html page and insert into the database.
 - iii) Display the all product details from the database on webpage.
 - iv) Delete a product based on product id.
 - v) Update product details based on product id.

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8. Create an application to verify the login.
 9. Create a simple application using sessions.
 10. Write a dynamic Servlet program which creates a connection to any database at runtime

JSP

1. Write a JSP Program to display the following output. Output: This is my first jsp.
2. Write a JSP Program which uses at least 6 implicit objects of jsp.
3. Write a JSP Program to create a database connection.
4. Create an application of product using JSP which has the following features
 - a. Html page to enter the product details (product id, product name, product price).
 - b. JSP to get the product details from the html page and insert into the database.
 - c. Display the all product details from the database on webpage
 - d. Delete a product based on product id.
 - e. Update product details based on product id.
5. Write a JSP Program using JSP taglib.

LIST OF IMPORTANT JAVA PROGRAMS

1. Java program to check given number is prime or not
2. Java program to print and count all the prime numbers present between given range?
3. Java program to print Fibonacci series up to given number?
4. Java program to check the number is palindrome or not
5. Java program to check the given number is Armstrong or not
6. Java program to convert number to word
7. Java program to check if the given number is autobiographical or not.

Note: The autobiographical number is a number whose sum of digits equal to number of digits.

example: 22100 ($2+2+1+0+0 == 5$ both are equal)

8. Java program to print nth prime number
9. Java Program to Display Alternate Prime Numbers
10. a) Java Program to Find Square Root of a Number Without sqrt Method
 - b) Java Program to Swap Two Numbers Using Bitwise Operator
 - c) Java Program to Find GCD of Two Numbers
 - d) Java Program to Find Largest of Three Numbers
 - e) Java Program to Find Smallest of Three Numbers Using Ternary Operator
 - f) Java Program to Check if a Number is Positive or Negative
 - g) Java Program to Check if a Given Number is Perfect Square
 - h) Java Program to Display Even Numbers From 1 to 100

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- i) Java Program to Display Odd Numbers From 1 to 100
 - j) Java Program to Find Sum of Natural Numbers
 - 11) Java Program to copy all elements of one array into another array
 - 12) Java Program to find the frequency of each element in the array
 - 13) Java Program to left rotate the elements of an array
 - 14) Java Program to print the duplicate elements of an array
 - 15) Java Program to print the elements of an array
 - 16) Java Program to print the elements of an array in reverse order
 - 17) Java Program to print the elements of an array present on even position and odd position
 - 18) Java Program to print the smallest element in an array
 - 19) Java Program to print the largest element and smallest element in an array
 - 20) Java Program to Print Odd and Even Numbers from an array
 - 21) Java program to Sort an Array in Java
 - 22) Java Program to print the sum of all the items of the array
 - 23) Java Program to right rotate the elements of an array
 - 24) Java Program to sort the elements of an array in ascending order and descending order
 - 25) Java Program to Find 2nd and 3rd and Largest Number in an array
 - 26) Java Program to Remove Duplicate Element in an array) Java program to count unique elements in an array
 - 27) 1) Java Matrix Programs
 - 28) Java Program to Add Two Matrices

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- 29) Java Program to Multiply Two Matrices
 - 30) Java Program to subtract the two matrices
 - 31) Java Program to determine whether two matrices are equal
 - 32) Java Program to display the lower triangular matrix and upper triangular matrix
 - 33) Java Program to find the frequency of odd & even numbers in the given matrix
 - 34) Java Program to find the product of two matrices
 - 35) Java Program to find the sum of each row and each column of a matrix
 - 36) Java Program to find the transpose of a given matrix
 - 37) Java Program to determine whether a given matrix is an identity matrix
 - 38) Java Program to determine whether a given matrix is a sparse matrix
 - 39) Java Program to count the total number of characters in a string
 - 40) Java Program to count the total number of punctuation characters exists in a String
 - 41) Java Program to count the total number of vowels and consonants in a string
 - 42) Java Program to determine whether two strings are the anagram
 - 43) Java Program to find all subsets of a string
 - 44) Java Program to find the longest repeating sequence in a string (very important)
 - 45) Java Program to find all the permutations of a string
 - 46) Java Program to replace lower-case characters with upper-case and vice-versa
 - 47) Java Program to find Reverse of the string and determine whether a given string is
palindrome
 - 48) Java Program to find maximum and minimum occurring character in a string
 - 49) Java program to find the duplicate characters in a string

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- 50) Java program to find the duplicate words in a string
 - 51) Java Program to find the frequency of characters
 - 52) Java Program to find the largest and smallest word in a string
 - 53) Java program to find longest even length word in a string
 - 54) Java Program to find the most repeated word in a text file
 - 55) Java Program to swap two string variables without using third or temp variable.
 - 56) Java Program to print smallest and biggest possible palindrome word in a given string
 - 57) Reverse String in Java Word by Word
 - 58) Reserve String without reverse () function
 - 59) Java program to convert given input from one form to another form (like decimal to octal/hexadecimal/binary)
 - 60) Java Program to create and display a singly linked list
 - (a) Java program to create a singly linked list of n nodes and count the number of nodes
 - (b) Java program to create a singly linked list of n nodes and display it in reverse order
 - (c) Java program to create a Circular Linked List of N nodes and count the number of nodes
 - (d) Java program to create a Circular Linked List of n nodes and display it in reverse order
 - (e) Java program to create a doubly linked list of n nodes and count the number of nodes
 - (f) Java program to create a doubly linked list of n nodes and display it in reverse order